Gaze Target Estimation with Concepts

Anonymous CVPR submission

Abstract

Estimating human gaze targets from images in-the-wild is an important and formidable task. Existing approaches primarily employ brittle, multi-stage pipelines that require explicit inputs, like head bounding boxes and human pose, in order to identify the subject of gaze analysis. As a result, detection errors can cascade and lead to failure. Moreover, these prior works lack the flexibility of specifying the gaze analysis task via natural language prompting, an approach which has been shown to have significant benefits in convenience and scalability for other image analysis tasks. To overcome these limitations, we introduce the ***Promptable Gaze Target Estimation (PGE)*** *task, a new end-to-end, concept-driven paradigm for gaze analysis. PGE conditions gaze prediction on flexible user text or visual prompts (e.g., "the boy in the red shirt" or "person in point [0.52, 0.48]") to identify a specific subject for gaze analysis. This approach integrates subject localization with gaze estimation, and eliminates the rigid dependency on intermediate analysis stages. We develop a scalable data engine to generate* ***Gaze-Co*** *(Gaze Estimation with Concepts), a dataset and benchmark of 120K high-quality, prompt-annotated image pairs. We also propose* ***AnyGaze****, the first model designed for PGE. AnyGaze uses a transformer-based detector to fuse features from frozen encoders and simultaneously solves subject localization, in/out-of-frame presence, and gaze target heatmap estimation. AnyGaze achieves state-of-the-art performance on multiple PGE benchmarks, setting a strong baseline for this new problem even on a difficult out-of-domain, real-world clinical dataset. We will open-source the AnyGaze model and the Gaze-Co benchmark.*

1. Introduction

Human gaze is a fundamental non-verbal cue, conveying a wealth of social and cognitive information [26, 44]. It is integral to human interaction, used to initiate social contact, signal attention and interest, manage conversational turn-taking, and regulate intimacy. As a direct proxy for cognition, gaze can also reveal a person's intentions, preferences, and emotional states [47, 55]. Consequently, its study has attracted significant interest across diverse fields, including psychology, human-computer interaction, and clinical research on conditions such as autism spectrum disorder (ASD) [1, 3, 20].

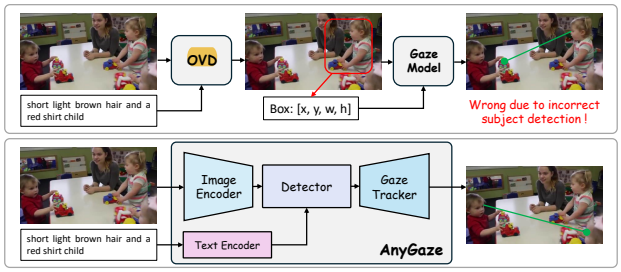

Figure 1. Gaze target estimation in in-the-wild environments. Prior methods such as Sharingan and ViTGaze have to rely on an Open-Vocabulary Detector (OVD) to produce auxiliary head boxes via dynamic human prompts, introducing a sequential dependency that becomes a major bottleneck.

Despite its importance, accurately estimating a person's gaze target in unconstrained, "in-the-wild" settings via image analysis remains a formidable challenge. Current approaches typically require explicit prior information (such as human head and face bounding boxes, pose estimations, and depth) not only for training but also as critical inputs during inference. Consequently, the most common architecture is a multi-stage pipeline where intermediate inputs are generated by a series of pre-processing steps prior to gaze estimation (Figure 1). This process typically requires first detecting and tracking a person and then precisely localizing their head or face with a bounding box before the gaze direction can be computed [10, 70]. This sequential dependency creates a critical bottleneck, as inaccuracies in these initial stages (common in crowded scenes, poor lighting, or with challenging cases like detecting children's faces) can cascade, leading to a failure of the entire system.

To overcome this limitation and provide greater convenience and flexibility in specifying gaze analysis tasks, we propose a paradigm shift toward an end-to-end, concept-based framework for gaze target estimation. This approach

is inspired by recent advancements in vision foundation models, such as the Open-vocabulary Detectors (OVDs) [23, 72], and the Segment Anything Model (SAM) series [2], which demonstrate remarkable abilities to detect and segment objects based on semantic-level text or visual concept prompts rather than explicit localization cues via bounding boxes. We extend this core idea to the domain of human gaze understanding, designing a promptable model that can directly identify the gaze target of a specified person within an image. By conditioning on a concept of the subject (e.g., "the boy in the red shirt"), our approach eliminates the dependencies on intermediate head bounding boxes or pose keypoints. This allows the model to infer gaze targets from a semantic understanding of the scene in a user-friendly end-to-end fashion, paving the way for more robust and versatile systems. Our main contributions can be summarized as:

(1) We define the **Promptable Gaze Target Estimation (PGE)** task, which extends the traditional gaze target estimation problem to an unconstrained, text promptable end-to-end paradigm.

(2) We design a scalable data engine to generate 120K high quality PGE training annotations consisting of subject text to gaze alignment data pairs.

(3) We introduce **AnyGaze**, the first promptable concept-driven gaze target estimation model. Our model achieves state-of-the-art performance in several benchmarks including an out-of-domain private dataset for autism children's gaze target estimation.

2. Related Work

Human Gaze Target Estimation. Interpreting gaze is a crucial component of human behavior understanding [19, 69]. This has motivated the "gaze-following" task, introduced by datasets like GazeFollow [47, 48], VideoAttentionTarget [13], GOO [57] and ChildPlay [54], which requires a model to predict the scene location a person is looking at. Dominant approaches have thus far employed multi-branch, fusion-based architectures. These models separately process explicit cues such as head position [51, 53, 65, 66], pose [5], text-based directions [62], facial expressions [33], and depth [27, 30, 58], subsequently combining these features to predict gaze points [9, 28, 29, 36, 40, 61, 68, 71]. In addition, existing end-to-end models cannot specify the subject person during inference either [16, 59, 61]. While effective, these strategies are dependent on the availability and accuracy of these intermediate representations. Concurrently, other studies have expanded the task's scope, such as multi-view gaze target estimation [41] and GazeHOI [55] for open-vocabulary targets. Despite these advances, a common limitation still persists: a dependency on auxiliary information, such as precise head bounding boxes or pose estimations. This reliance poses significant challenges in realistic settings, where such priors are often unreliable or unavailable.

Promptable and Interactive Object Detection. Recent advances in promptable and open-vocabulary perception are enabling models to generalize beyond fixed label sets [7, 43]. OVD methods, for instance, leverage large-scale vision-language encoders like CLIP to detect arbitrary concepts specified by text at inference time, even for categories unseen during training [23, 37, 42, 74]. In parallel, interactive segmentation frameworks such as SAM 3 [2] demonstrate how models can respond to flexible text and visual prompts. Together, these advances suggest the new paradigm: visual perception systems can be effectively guided by conceptual cues, rather than rigidly predefined supervision. Building on this paradigm, we formulate gaze target estimation as a concept-conditioned reasoning task, where both the subject and their attended region are inferred from semantic prompts rather than explicit localization inputs.

Vision Foundation Models. Vision Foundation Models (VFMs) have become a dominant approach in computer vision, entailing a significant shift of models trained on massive web-scale datasets. VFMs primarily include two branches: (1) weakly-supervised models like CLIP [45], SigLIP [60, 67], and MetaCLIP [6, 14, 63], which learn powerful representations from image-text pairs using contrastive losses, and (2) self-supervised learning (SSL) models like DINO series [31, 31, 52], which learn robust visual features from unlabeled images. These powerful, pre-trained encoders now serve as general-purpose backbones for a wide range of downstream tasks. This trend has also influenced gaze estimation, where systems like ViTGaze [53] and Gaze-LLE [51] have successfully adapted VFM architectures and leveraged their pre-trained features to improve performance, demonstrating the value of these models for fine-grained, human-centric tasks.

3. Method

3.1. Promptable Gaze Target Estimation

We define the PGE task as estimating the gaze target location of a specific subject within an image or video, identified by a user-provided prompt. This prompt can be of two types: text prompting via a short, natural language query; and visual one by prompting a spatial coordinate, such as the center point of a head bounding box. In text prompting, our goal is to support any simple, visually-groundable noun phrase as a text prompt. However, this introduces intrinsic ambiguity (e.g., "the person in the back"). To mitigate this and enable unambiguous identification, we structure text prompts around four main categories. A user can combine descriptions from these categories to specify a subject: (1) Appearance: Noun phrases describing a person, consisting of an identity (e.g., woman, man, child) and optional modifiers

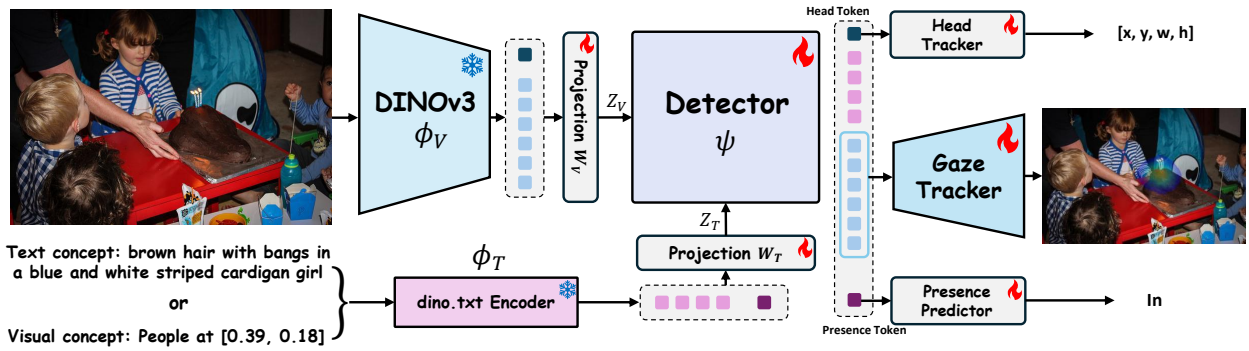

Figure 2. An overview of the AnyGaze end-to-end framework for PGE. The model uses frozen image (DINOv3) and text (dino.txt) encoders to provide features to a trainable transformer-based detector. This detector utilizes multiple decoders to simultaneously predict a subject's head bounding box , in-frame presence , and gaze target. AnyGaze is conditioned on flexible user prompts, such as a natural language text description or visual (coordinate-based) cues.

(e.g., hair type/color, clothes, glasses, hat). (2) Location: The subject's spatial position in the image (e.g., center, left, top-right). (3) Pose: The subject's static posture. (4) Action: Verb phrases describing what the subject is doing.

More formally, given an input RGB image $I \in \mathbb{R}^{3\times H\times W}$ and a prompt P (either text T or a visual cue), the goal of PGE is to produce a gaze heatmap $\hat{H} \in \mathbb{R}^{H_{\text{out}}\times W_{\text{out}}}$. Each element $\hat{H}(i,j)$ represents the probability that the subject specified by P is gazing at the spatial location (i,j). Unlike classic gaze estimation methods, PGE demands that the model solve the task end-to-end, directly linking a flexible, high-level query to a final gaze heatmap. This formulation is substantially more challenging as it precludes the use of auxiliary inputs common in traditional pipelines (e.g., subject bounding boxes, pose keypoints, or depth maps). Our model must implicitly learn to perform subject identification, localization, and gaze estimation jointly with text input, rather than relying on the explicit outputs of separate, specialized models like open-vocabulary detectors or pose estimators.

3.2. AnyGaze Architecture

Figure 2 shows the overall architecture of AnyGaze. The model consists of a frozen image encoder, a frozen text encoder to proceed visual modality and text modality. A transformer-based detector is used to learn joint representations and map text prompt into the main gaze target estimation and auxiliary tasks.

Image Encoder. We use a frozen ViT, denoted $\phi_V(\cdot)$, as our image encoder to extract general visual features. Consistent with PGE's problem definition, we do not employ any auxiliary models for dedicated depth or pose feature extraction. The image encoder processes an input image $I \in \mathbb{R}^{3\times H\times W}$ by dividing it into a sequence of N_V patch tokens, to which a learnable `[CLS]` token is prepended. The resulting output sequence from ϕ_V is $\phi_V(I) = [c, s_1, s_2, \ldots, s_{N_V}] \in \mathbb{R}^{(1+N_V)\times D_V}$. Here, D_V is the embedding dimension, $c \in \mathbb{R}^{D_V}$ is the final `[CLS]` token embedding, and $s_i \in \mathbb{R}^{D_V}$ is the output embedding for the i-th patch.

Text Encoder. We employ a frozen text encoder, $\phi_T(\cdot)$, which consists of a series of transformer blocks and a final linear layer. The linear layer maps the output `[EOS]` token's feature to the image embedding space. To prepare the input, a tokenizer first converts the text T into a sequence of token IDs. These IDs are then mapped to initial text embeddings T_E via an embedding layer, and the sequence is padded to a fixed context length, L_T. The encoder ϕ_T processes this embedding sequence T_E, producing the final output sequence: $\phi_T(T_E) = [t_1, t_2, \ldots, t_{N_T}, t_{eos}, \ldots, t_{pad}] \in \mathbb{R}^{L_T\times D_T}$. Here, D_T is the output embedding dimension, $t_i \in \mathbb{R}^{D_T}$ is the output embedding for the i-th content token, and $t_{eos}, t_{pad} \in \mathbb{R}^{D_T}$ are the special tokens representing the end of the input and padding, respectively.

Projection Layers. The image encoder ϕ_V and text encoder ϕ_T output features with potentially different dimensions, D_V and D_T, respectively. To map these features into a unified, shared space, we introduce two trainable linear projection layers, W_V and W_T. These layers project the high-dimensional features into a common, lower-dimension D, where $D < \min(D_V, D_T)$:

$$Z_{\text{V}} = W_V \cdot \phi_V(I) \tag{1}$$

$$Z_{\text{T}} = W_T \cdot \phi_T(T_E) \tag{2}$$

Here, W_V and W_T are the learnable projection matrices, and the operation $\cdot$ denotes a per-token linear transformation. This process results in a sequence of visual tokens $Z_{\text{V}} \in \mathbb{R}^{(1+N_V)\times D}$ and text tokens $Z_{\text{T}} \in \mathbb{R}^{L_T\times D}$, which now share the same embedding dimension.

Task-Specific Embeddings. Beyond the primary cross-modal feature alignment, we introduce two specialized, learnable embeddings to explicitly model key sub-problems: a **head token** and a **gaze presence token**.

1. **Head Token:** This is a learnable embedding designed to explicitly predict the head localization of the prompted subject, serving as the image-text alignment objective in our task. It is initialized using the embedding of the text `[EOS]` token.
2. **Target Presence Token:** This token is introduced to address the *in/out-of-frame* gaze target boolean prediction objective. The rationale for this is that the in/out decision relies on global contextual cues from the entire image, which conflicts with the inherently local nature of the target localization objective. Forcing a single query or mechanism to handle both can be counterproductive. Therefore, we decouple the localization and in/out prediction tasks. This dedicated, learnable global token is responsible for the in/out prediction and is initialized using the embedding of the visual `[CLS]` token.

Detector Transformer. After extracting and projecting the visual and text features, we introduce a **Detector Transformer**, $\psi(\cdot)$, to fuse these representations and refine them for the gaze target estimation task. $\psi(\cdot)$ apply the same Transfomrer block in DINOv3 [52]. The input to ψ is a single sequence F constructed by concatenating the projected features and our specialized task tokens.

First, we define the **head token** $t_h \in \mathbb{R}^D$ and **target presence token** $t_p \in \mathbb{R}^D$. These are formed by combining the projected global tokens (c' from vision, t'_{eos} from text) with dedicated learnable embeddings, E_{head} and $\mathrm{E}_{\text{presence}}$:

$$t_h = t'_{eos} + \mathrm{E}_{\text{head}} \tag{3}$$

$$t_p = c' + \mathrm{E}_{\text{presence}} \tag{4}$$

Let $\mathbf{s}' = [s'_1, \ldots, s'_{N_V}]$ be the sequence of projected visual patch tokens from Z_V (excluding c') and $\mathbf{t}' = [t'_1, \ldots, t'_{N_T}]$ be the projected text content tokens from Z_T (excluding t'_{eos} and padding). The full input sequence F is then assembled as:

$$F = [t_h, \mathbf{t}', \mathbf{s}', t_p] \in \mathbb{R}^{(N_T+N_V+2)\times D} \tag{5}$$

We inject positional information by adding 1D sinusoidal position embeddings to the text tokens $\mathbf{t}'$ and 2D sinusoidal position embeddings to the visual tokens $\mathbf{s}'$ [18]. The Detector transformer ψ is a stack of k standard transformer blocks; k is a hyperparameter ablated in our experiments. ψ processes F and outputs a refined sequence of the same dimension, $\psi(F) \in \mathbb{R}^{(N_T+N_V+2)\times D}$. Specific tokens from this output are then passed to dedicated decoders.

Decoders. The Detector transformer ψ outputs a refined sequence of tokens. We attach three distinct prediction heads to specific tokens from this sequence to produce the final outputs.

- **Gaze Tracker (Heatmap Decoder):** The refined visual patch tokens $\hat{\mathbf{s}} \in \mathbb{R}^{N_V\times D}$ are first re-assembled from their 1D sequence form back into a 2D spatial grid. This feature map is then fed through a convolutional decoder, consisting of two transposed convolutional layers, which upsamples the features to the output heatmap $\hat{H} \in \mathbb{R}^{H_{out}\times W_{out}}$. In our experiments, we set $H_{out} = W_{out} = 128$.
- **Head Tracker (Box Decoder):** We use the refined head token $\hat{t}_h \in \mathbb{R}^D$ for an auxiliary head localization task. The token is passed through a 3-layer feed-forward network (FFN) with ReLU activations and a hidden dimension of D. This head regresses a 4-dimensional vector $[x, y, w, h]$ representing the normalized center coordinates, width, and height of the subject's head box.
- **Presence Predictor (In/Out Decoder):** The refined gaze presence token $\hat{t}_p \in \mathbb{R}^D$ is used to predict whether the gaze target is in or out of the frame. It is processed by a 2-layer FFN (with one hidden layer of dimension D and ReLU activation) that outputs a single logit for the binary classification.

3.3. Learning Objective

We train our model end-to-end with a joint multi-task objective. The total loss $\mathcal{L}_{\text{total}}$ is a weighted linear combination of three loss terms: one for the gaze heatmap, one for the gaze presence, and one for the auxiliary head localization task.

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{gaze}} + \mathcal{L}_{\text{presence}} + \mathcal{L}_{\text{head}} \tag{6}$$

The gaze heatmap loss $\mathcal{L}_{\text{gaze}}$ is a pixel-wise binary cross-entropy (BCE) loss. The supervisory target is a heatmap Y, constructed by placing a 2D Gaussian ($\sigma = 3$) at the ground-truth gaze target location. Let $\hat{Y}$ be the predicted heatmap. The loss is defined as:

$$\mathcal{L}_{\text{gaze}} = -\frac{1}{N}\sum_{p=1}^{N}\left[y_p \log(\hat{y}_p) + (1-y_p)\log(1-\hat{y}_p)\right] \tag{7}$$

where $N = H_{out} \times W_{out}$ is the total number of pixels, and y_p and $\hat{y}_p$ are the ground-truth and predicted values for a single pixel p, respectively.

The gaze presence loss $\mathcal{L}_{\text{presence}}$ is a Focal Loss supervised with a binary label $Y_{\text{presence}} \in \{0, 1\}$. Let $\hat{Y}_{\text{presence}} \in [0, 1]$ be the model's predicted probability that the target is present ($Y_{\text{presence}} = 1$). The loss is defined as:

$$\mathcal{L}_{\text{presence}} = \mathcal{L}_{\text{focal}}(Y_{\text{presence}}, \hat{Y}_{\text{presence}}) \tag{8}$$

where the hyperparameter of the focal loss is the default value from [35].

The subject head bounding box loss $\mathcal{L}_{\text{head}}$ is a linear combination of the $\mathcal{L}_1$ loss and the generalized IoU loss. which is widely used by object detection tasks. It defined as:

$$\mathcal{L}_{\text{head}} = \lambda_{l_1} ||b - \hat{b}||_1 + \lambda_{\text{iou}} \mathcal{L}_{\text{iou}}(b, \hat{b}) \tag{9}$$

where b and $\hat{b}$ is the ground truth head box and predicted head box. $\mathcal{L}_{\text{IoU}}$ is the GIoU loss [7, 50]. λ_{l_1} and λ_{iou} are head object detection hyperparameters. We followed DETR [7] and OWLViT [42] to set $\lambda_{l_1} = 5$ and $\lambda_{\text{iou}} = 2$.

4. Gaze with Concept (Gaze-Co) Dataset

Training AnyGaze for the PGE task requires a large and diverse dataset annotated with concepts, a resource that no existing gaze dataset provides. To address this, we developed a scalable data engine that generates annotations via a human-in-the-loop feedback process. This engine worked in tandem with two human annotators (co-authors) to perform several key functions: aligning heterogeneous annotations, filtering low-quality frames, generating concise concept phrases, and facilitating human verification. After three rounds of iteration, we created Gaze-Co, the first large-scale dataset for PGE, containing 120K samples sourced from the training set of GazeFollow, VisualAttentionTarget (VAT) and ChildPlay. To establish a comprehensive benchmark, we also converted the test sets of these well-known gaze datasets to the PGE format, creating GazeFollow-Concept, VAT-Concept, and ChildPlay-Concept. We further conducted experiments on a private, Institutional Review Board (IRB)-approved, out-of-domain (OOD) evaluation set with 40 child social communication (Child-SC) videos.

4.1. Data Engine

Figure 3 illustrates the workflow of the data engine. We can divide the process into three stages: (1) data alignment and filter; (2) concept generation; (3) verification.

Data Alignment and Filter. The source datasets differ in coordinate conventions, split policies, and metadata. We therefore adopt a unified schema with explicit pixel coordinates for the head box $(x_{min}, y_{min}, x_{max}, y_{max})$, and a normalized gaze point $(g_x/W, g_y/H)$. Then, to ensure reliable person-scale learning signals, we apply geometric and sharpness filters. Annotations are retained only if the head box width ≥ 30 px, height ≥ 40 px, area ≥ 2500 px^2, and the box-to-image ratio $\in [0.008, 0.3]$, with sufficient Tenengrad focus. These thresholds remove extremely small, oversized, or blurry instances while preserving diverse valid samples.

Concept Generation. For each retained annotation, we produce a short, lowercase concept phrase comprising attribute, position, action, and pose, together with a coarse count of visible people. Concept generation is executed with a production Vision Language Model (VLM) accessed through API (Gemini 2.5 Pro [15, 24]), using batch processing with

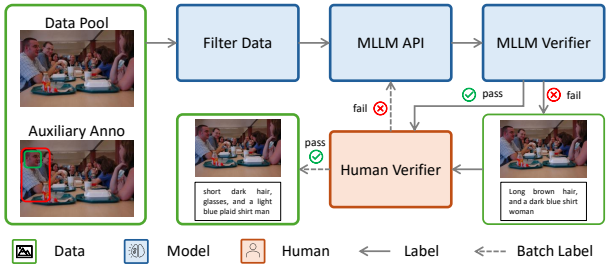

Figure 3. Overview of the AnyGaze data engine.

fixed prompts and rates. The attribute phase privileges stable visual cues (hair, glasses, beard, colors, and patterns) and the final token is constrained to one of man, woman, boy, girl, infant, child as an apparent (perceived) age/sex presentation label, used solely as a visual category cue rather than a verified identity attribute; when indeterminate, we write "adult" or "child. The position uses brief canvas references (e.g., "bottom left corner"). Action and pose are explicitly non-overlapping: action describes ongoing interaction or motion with object or direction when visible, while pose captures static body configuration and facing direction. When a field is indeterminate, we write "none."

Verification. We adopt an Multi-modal Large Language Model (MLLM)-first, human-in-the-loop verification workflow. The Gemini 2.5 Pro reviews all generated concepts and flags each as pass or fail. Human annotators then spot-check a random subset of the MLLM passed cases and evaluate the batch success rate. During review, both the MLLM and human annotators check whether each concept correctly matches its designated head box (consistency), whether all four fields are present and non-conflicting (completeness), and whether the text contains no sensitive or identifying information (privacy). If the human verifier finds the batch success rate is low, the data engine will return to the concept generation stage. The human verifier then adjusts the prompts and rules and re-runs the concept generation and verification until the observed error rate is kept low ($\leq 1\%$). For the private Child-SC dataset with IRB restriction, all concept annotations are generated manually by authorized human annotators without being sent to the MLLM.

4.2. Gaze-Co Dataset and Benchmarks

Gaze-Co is the first large-scale dataset for promptable gaze target estimation, unifying GazeFollow, VAT, and ChildPlay under a shared schema with concept-level annotations for both training and evaluation.

Training Data. The Gaze-Co 120K training set contains about 120K images from the official training splits of the three source datasets. Each record includes the target head box, normalized gaze point, in/out-of-frame label, and a compact concept phrase (attribute, position, action, and pose). All samples pass image quality filters, ensuring diverse, valid

instances across viewpoint, poses, scales, and interaction contexts.

Benchmark Settings. The benchmark uses the official test splits of GazeFollow, VAT, and ChildPlay, each converted into the Gaze-Co format. We evaluate concept-conditioned gaze prediction under three settings: (i) in-domain testing on the test sets; and (ii) OOD evaluation on the Child-SC dataset, a developmental sample of children's gaze behavior collected under an IRB-approved study (see the appendix for the dataset description). Every text prompt in the test set has been human-verified rather than spot-checked, ensuring accuracy and consistency. Each model receives the image, with the concept text added or altered according to the test setting. This setup provides a consistent framework for comparing models under controlled concepts-based conditions.

4.3. Metrics

We evaluate models using heatmap Area Under the Curve (AUC) and pixelwise L2. For heatmap AUC, the predicted heatmap is treated as a confidence map to compute an ROC curve against the binary gaze target map. Pixelwise L2 measures the Euclidean distance between the heatmap peak and the ground-truth gaze point. For GazeFollow-Concept, each image includes multiple gaze annotations directed at the same target person, so we additionally report Avg L2 (distance to the mean of all targets) and Min L2 (distance to the nearest target). For VAT-Concept, ChildPlay-Concept, and Child-SC (the IRB-approved OOD set), annotations include binary in/out labels relative to the target region; thus, we report region-based AUC, pixelwise L2, and average precision (AP) to jointly evaluate localization and in/out binary classification.

5. Experiments

We evaluate AnyGaze on the PGE task, comparing its text-prompting capabilities against State-of-the-Art (SOTA) two-stage pipelines that integrate OVDs for head/human detection with a separate gaze modeling stage. We also present a series of ablation studies demonstrating the importance of the frozen encoders, validating our loss design, and analyzing the differences between visual and text prompting. Finally, we demonstrate a real-world application, the "AnyGaze Agent," a system that integrates AnyGaze with an Augmented Reality (AR) device and a MLLM.

5.1. Implementation Details

All models are trained for 25 epochs using the Adam optimizer and a cosine learning rate scheduler with an initial rate of 1e-3 and batch size 128, followed by an additional 5 epochs with a reduced learning rate of 1e-5. All training experiments are conducted with 4 NVIDIA H100 GPUs. The inference is running with 1 NVIDIA L40S GPU. We adopt the DigiLens ARGO smartglass as the AR platform to deploy AnyGaze Agent for real world experiments. More details are shown in the appendix.

5.2. Main Results

Table 1 compares AnyGaze against strong two-stage baselines, which we created by pairing three SOTA gaze methods Gaze-LLE [51], Sharingan [56], ViTGaze [53] with four leading OVDs for human detection [23, 37, 43, 74]. Details of these baselines are shown in the appendix. The encoders of AnyGaze can be CLIP-L [45] or DINOv3-L [52] with dino.txt [31]. On the PGE text-prompting task, AnyGaze achieves SOTA performance on all metrics across the three public datasets, as well as on our challenging OOD private dataset from a real-world assessment setting in which children's social communication skills are quantified by experts.

5.3. Ablation Study

We use AnyGaze-DINOv3-L for all following up ablation experiments in GazeFollow-Concept and VAT-Concept.

Compare AnyGaze with SOTA VLMs. To demonstrate the utility of AnyGaze in PGE, we evaluate the 0-shot performance of SOTA VLM on gaze point prediction. Table 2 shows the comparison of AnyGaze, Qwen3-VL-8B and Gemini 2.5 Flash. AnyGaze surpass all of them, highlighting the importance of building specific model for PGE.

PGE with Different Prompting. AnyGaze supports both visual (coordinate-based text) and text (natural language) prompts, as illustrated in Figure 2. In Table 3, we compare the performance of these different strategies. We find that text-based prompting achieves performance on par with visual prompting. Furthermore, our decomposition analysis of text prompt composition reveals that the subject's appearance and pose description are the most critical components for the PGE task.

Impact of Frozen Encoder. A key design choice for AnyGaze is to keep the image and text encoders frozen. We validate this approach in Table 4, which compares the default frozen model against one where the DINOv3 image encoder or the text encoder are fine-tuned. Unfreezing image or text encoders leads to a clear drop in performance. This demonstrates that DINOv3's pre-trained features are highly robust and generalizable for the PGE task, and that fine-tuning may lead to overfitting or harmful feature drift.

Loss ablations. We conducted an ablation study (Table 5) on our objective function's components: gaze heatmap, presence, and head losses. The essential gaze heatmap loss was always active, while we trained models removing the presence loss, the head loss, and both. Results indicate the presence loss only supports the auxiliary in/out prediction, not help gaze estimation. The head loss, however, improves both the gaze target estimation and the target presence prediction.

Impact of Detector Dimension. We study the impact of the Detector transformer's layer dimension D in Table 6. The

Gaze Model	OVD Head Detection	Param	Time[ms]	GazeFollow-Concept			VAT-Concept			ChildPlay-Concept			Child-SC (OOD)		
				AUC ↑	Avg L2 ↓	Min L2 ↓	AUC ↑	L2 ↓	AP ↑	AUC ↑	L2 ↓	AP ↑	AUC ↑	L2 ↓	AP ↑
ViTGaze [53]	Detic [74]	65M	53	0.869	0.138	0.216	0.810	0.407	0.700	0.895	0.201	0.878	0.892	0.174	0.772
	GroundingDINO-B [37]	255M	116	0.913	0.116	0.192	0.906	0.232	0.792	0.919	0.170	0.863	0.925	0.139	0.795
	LLMDet-L [23]	365M	339	0.897	0.113	0.189	0.820	0.235	0.763	0.910	0.158	0.869	0.891	0.165	0.825
	OWLv2-L [43]	460M	179	0.927	0.094	0.170	0.846	0.241	0.749	0.945	0.151	0.862	0.878	0.168	0.781
Sharingan [56]	Detic [74]	149M	57	0.849	0.279	0.197	0.849	0.330	0.746	0.897	0.281	0.873	0.836	0.355	0.849
	GroundingDINO-B [37]	338M	119	0.860	0.259	0.178	0.865	0.352	0.720	0.916	0.216	0.888	0.867	0.349	0.771
	LLMDet-L [23]	449M	342	0.865	0.241	0.160	0.866	0.281	0.785	0.914	0.213	0.874	0.851	0.347	0.832
	OWLv2-L [43]	544M	182	0.867	0.255	0.173	0.877	0.291	0.745	0.936	0.185	0.893	0.941	0.263	0.847
Gaze-LLE [51]	Detic [74]	350M	83	0.933	0.179	0.117	0.842	0.239	0.787	0.917	0.146	0.894	0.886	0.198	0.843
	GroundingDINO-B [37]	539M	145	0.925	0.165	0.106	0.817	0.234	0.780	0.910	0.152	0.863	0.911	0.172	0.856
	LLMDet-L [23]	650M	368	0.912	0.168	0.109	0.832	0.230	0.793	0.915	0.145	0.875	0.912	0.175	0.855
	OWLv2-L [43]	745M	208	0.941	0.146	0.089	0.847	0.229	0.792	0.931	0.127	0.893	0.912	0.161	0.830
AnyGaze-CLIP-L		430M	35	0.953	0.105	0.056	0.913	0.137	0.874	0.951	0.104	**0.915**	0.928	0.146	0.868
AnyGaze-DINOv3-L		870M	96	**0.958**	**0.099**	**0.050**	**0.928**	**0.123**	**0.879**	**0.954**	**0.098**	0.906	**0.949**	**0.090**	**0.902**

Table 1. PGE results on four datasets. The input is the text prompt of the subject person's appearance, position, action and pose. For baseline methods, the OVD is used to extract the bounding box with the input prompt and then feed the bounding box to the gaze models. Latency is compared with inference running speed per image in *batch size* = 1.

Model	Param	GazeFollow-Concept		VAT-Concept	
		Avg L2 ↓	Min L2 ↓	L2 ↓	AP ↑
Qwen3-VL-8B [4]	8B	0.201	0.137	0.286	0.651
Gemini 2.5 Flash [15]	-	0.216	0.156	0.292	0.661
AnyGaze-DINOv3-L	870M	**0.099**	**0.050**	**0.123**	**0.879**

Table 2. Compare AnyGaze with SOTA VLMs.

Prompt type	GazeFollow-Concept			VAT-Concept		
	AUC ↑	Avg L2 ↓	Min L2 ↓	AUC ↑	L2 ↓	AP ↑
No prompting	0.944	0.144	0.090	0.875	0.210	0.796
visual prompting	0.958	0.100	0.050	0.914	0.131	0.894
text prompting (appearance)	0.952	0.113	0.062	0.904	0.153	0.840
text prompting (position)	0.953	0.124	0.073	0.893	0.188	0.826
text prompting (action)	0.949	0.129	0.078	0.897	0.180	0.839
text prompting (pose)	0.952	0.121	0.070	0.909	0.163	0.859
text prompting (all)	0.958	0.099	0.050	0.928	0.123	0.879

Table 3. Comparison of different prompt strategies.

Visual	Text	Trainable Param	GazeFollow-Concept			VAT-Concept		
			AUC ↑	Avg L2 ↓	Min L2 ↓	AUC ↑	L2 ↓	AP ↑
✗	✗	870.2M	0.931	0.150	0.095	0.886	0.212	0.823
✓	✗	332.1M	0.943	0.119	0.067	0.886	0.206	0.813
✗	✓	541.7M	0.952	0.130	0.078	0.874	0.201	0.825
✓	✓	3.6M	**0.958**	**0.099**	**0.050**	**0.928**	**0.123**	**0.879**

Table 4. Comparison of the encoder frozen strategies.

$\mathcal{L}_{gaze}$	$\mathcal{L}_{presence}$	$\mathcal{L}_{head}$	GazeFollow-Concept			VAT-Concept		
			AUC ↑	Avg L2 ↓	Min L2 ↓	AUC ↑	L2 ↓	AP ↑
✓	✗	✗	0.956	0.102	0.052	0.925	0.135	-
✓	✓	✗	0.955	0.103	0.055	0.924	0.136	0.863
✓	✗	✓	0.958	0.099	0.051	0.924	0.128	-
✓	✓	✓	**0.958**	**0.099**	**0.050**	**0.928**	**0.123**	**0.879**

Table 5. Ablation experiment on loss selection.

results indicate that performance plateaus at $D = 128$. We observed no significant performance gain from increasing D further, and thus selected $D = 256$ as it provides the best trade-off between accuracy and computational cost.

D of $\psi(\cdot)$	GazeFollow-Concept			VAT-Concept		
	AUC ↑	Avg L2 ↓	Min L2 ↓	AUC ↑	L2 ↓	AP ↑
64	0.953	0.115	0.062	0.915	0.144	0.871
128	0.960	0.100	0.050	0.928	0.116	0.875
256	0.958	0.099	0.050	0.928	0.123	0.879
512	0.959	0.104	0.054	0.917	0.122	0.875

Table 6. Ablation experiment on the selection of the Detector Transformer dimension.

5.4. Visualization

Figure 4 showcases qualitative gaze estimation results from AnyGaze. The input, displayed in the black boxes, is a text prompt describing only the subject's appearance, such as "light brown hair and a blue striped shirt boy". The visualizations demonstrate that AnyGaze performs robustly not only in simple scenarios with 2-3 people but also in complex, crowded scenes with four or more individuals. Notably, the final two examples, "long black high ponytail hair and a pink shirt girl" and "short blonde hair wearing a dark blue and light gray shirt boy", are from an OOD Child-SC video dataset. The model's successful performance on this unseen data highlights its generalization and robustness.

5.5. AnyGaze Agent in Augmented Reality (AR)

Previous gaze estimation models ignore the real-world application experiment. Inspired by recent tool-enhanced MLLM workflows [2, 64], we developed the AnyGaze Agent (Figure 5). This system uses a central MLLM (Gemini 2.5 [15]) that leverages AnyGaze as a specialized tool to solve advanced user queries, such as, "How many gaze shifts does this girl with the white dress present?". The workflow captures User Audio and Environment Images from an AR Glass. MLLM calls Whisper v3 [46] to transcribe the audio, followed by query reasoning and prompt rephrasing. The agent converts the high-level query into a low-level text prompt

Figure 4. Visualization of AnyGaze's gaze target estimation results from several datasets. The input is the text prompt (shown in the black box) describing only the subject person's appearance and the image. AnyGaze detect the subject's head and track the gaze target. More qualitative comparison is shown in the appendix.

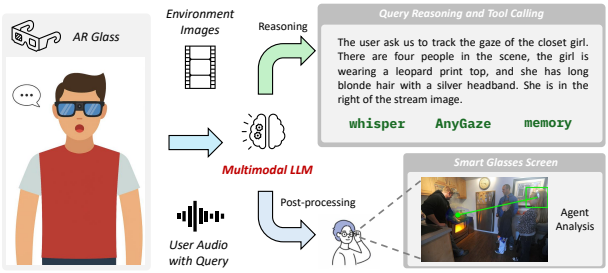

Figure 5. Workflow of MLLM-powered AnyGaze Agent.

MLLM	Gaze Shift MAE/min ↓	Eye Contact MAE/min ↓
Gemini 2.5 Flash (Raw)	9.247	14.763
Gemini 2.5 Flash (AnyGaze Agent)	2.337	4.756
Gemini 2.5 Pro (Raw)	6.063	10.268
Gemini 2.5 Pro (AnyGaze Agent)	2.546	6.672

Table 7. Comparison of AnyGaze Agent and no agent framework.

(e.g., "girl with white dress") , calls the AnyGaze tool to generate gaze tracing, and then uses its VLM function to analyze the post-processed video, providing the user with the required analysis.

We collected 10 real-world videos with rich gaze movement using the DigiLens ARGO AR glass to test the agent's performance. The evaluation focused on two tasks: gaze shift calculation and eye contact calculation with other social partners. Using the Mean Absolute Error (MAE) per-minute, the AnyGaze Agent demonstrated significantly better performance than a raw, single MLLM solution. Results of AnyGaze Agent experiment is showed in Table 7.

6. Discussion

The practical applications of human gaze target estimation are diverse and impactful. In healthcare, for instance, this technology can significantly enhance the analysis of non-verbal communication behaviors which are implicated in the diagnosis and treatment of developmental conditions such as autism [49]. In order for AI models to be used in clinical applications, they must be sufficiently robust and easy to use by nonexperts. This works takes a significant step in that direction for the task of gaze assessment. Our concept-based approach, which allows subjects to be identified by their attributes in natural language, is a first step towards the flexible and convenient specification of a broad set of behavioral analysis tasks. In addition, by creating a unified end-to-end learnable architecture we increase robustness by eliminating brittle stage-wise approaches to identifying the subjects of gaze analysis. Our approach is beneficial even in comparison to using state-of-the-art OVD models to identify subjects, e.g. the SOTA OVD OWLv2 has only a 70% detection accurate rate in Child-SC for the child head and face detection tasks.

7. Conclusion

We present AnyGaze, a system that enables interactive human gaze target estimation using flexible, open-vocabulary text prompts to identify the subject. Our principal contributions include introducing the novel Promptable Gaze Target Estimation (PGE) task and Gaze-Co benchmark, proposing a tailored transformer-based detector and learning objective, and developing a human-and-AI-in-the-loop data engine to adapt existing datasets. AnyGaze achieves state-of-the-art results in Gaze-Co benchmark, and its robustness is further validated on a challenging out-of-domain (OOD) dataset of child social communication videos. We believe AnyGaze and the Gaze-Co benchmark represent important milestones, paving the way for future research and applications in social AI and human behavior understanding.

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